



Sydney Girls High School

--	--	--	--	--	--	--	--

Student Number

2023

YEAR 11

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided on pages 13 -15
- Write your student number on the multiple choice answer sheet and the cover page

Total marks – 35

Part A – Pages 2-5

10 marks

- Attempt Questions 1 – 10

Part B – Pages 6-8

13 marks

- Attempt Questions 11 – 12

Part B – Pages 9-11

12 marks

- Attempt Questions 13 – 15

Part A – 10 marks

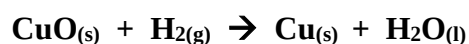
Attempt Questions 1 – 10

Use the multiple-choice answer sheet for Questions 1–10

- 1 Which observation will be made when a clean strip of magnesium is dipped into aqueous copper sulfate:

- (A) Hydrogen gas is released.
- (B) Sulfur dioxide gas is released.
- (C) The blue colour of the copper nitrate becomes more intense.
- (D) The magnesium strip becomes coated with a reddish deposit of copper.

- 2 Hydrogen and copper (II) oxide react to produce metal in water.



Which row correctly describes the oxidising and reducing agent?

	Oxidising agent	Reducing agent
(A)	H ₂ O	Cu
(B)	H ₂	CuO
(C)	Cu	H ₂ O
(D)	CuO	H ₂

- 3 Which combination of the following periodic table trends would result in an increase in a metal's reactivity?

- (A) Decrease in atomic radius, decrease in electronegativity.
- (B) Increase in ionisation energy, increase in atomic radius.
- (C) Increase in ionisation energy, decrease in electronegativity.
- (D) Decrease in ionisation energy, decrease in electronegativity.

4 Which equation represents a neutralisation reaction?

- (A) $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(l)}$
- (B) $\text{C}_6\text{H}_{12}\text{O}_{6(aq)} + 6\text{O}_{2(g)} \rightarrow 6\text{CO}_{2(g)} + 6\text{H}_2\text{O}_{(g)}$
- (C) $\text{CuCl}_{2(aq)} + 2\text{NaOH}_{(aq)} \rightarrow \text{Cu}(\text{OH})_{2(s)} + 2\text{NaCl}_{(aq)}$
- (D) $2\text{HNO}_{3(l)} + \text{Ba}(\text{OH})_{2(aq)} \rightarrow \text{Ba}(\text{NO}_3)_{2(aq)} + 2\text{H}_2\text{O}_{(l)}$

5 Which of the following is a true statement that can be used to predict the entropy change in a system?

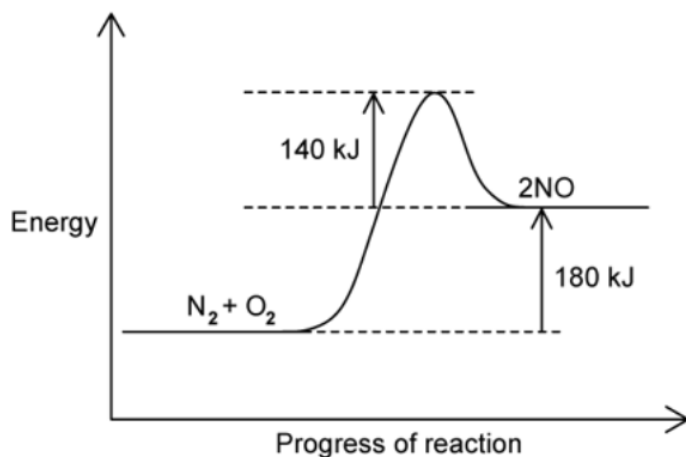
- (A) As the temperature of a substance increase, the entropy increases.
- (B) A substance in its gaseous state has less entropy, than the same amount of the substance in its liquid state.
- (C) A substance in its liquid state has less entropy than the same amount of substance in its solid state.
- (D) When a solute dissolves in a solvent, the solute particles in solution can be arranged in fewer ways than when they were solid, so the amount of entropy usually decreases

6 Catalysts such as iron and nickel are commonly used in industrial reactions.

Which statement about a catalyst is NOT correct?

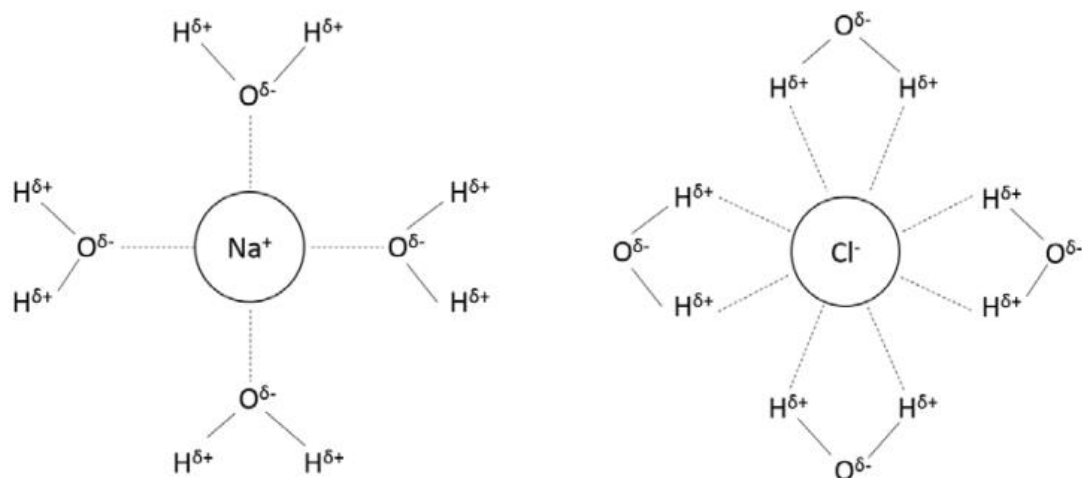
- (A) They create a pathway which has a lower activation energy.
- (B) Their concentration does not change throughout the reaction.
- (C) They increase the amount of energy required for a reaction to occur.
- (D) They can allow reaction to occur at much lower temperatures.

- 7 What is the activation energy for this reaction?



- (A) +140 kJ
- (B) +180 kJ
- (C) +320 kJ
- (D) -320 kJ
- 8 Which of the following is a statement describing Hess' Law?
- (A) Energy cannot be created nor destroyed, but it can be changed from one form into another.
- (B) In a chemical reaction, the total mass of the reactants is equal to the total mass of the products.
- (C) The enthalpy change for a chemical reaction is the same regardless of the number of steps involved in the process.
- (D) Enthalpy change in a chemical reaction equals the sum of the enthalpy of all products minus the sum of enthalpies of all reactants.

- 9 Diagram and models are often used to demonstrate complex chemical ideas.



Which chemical process is demonstrated by these diagrams?

- (A) Ionisation.
 - (B) Carbonation.
 - (C) Precipitation.
 - (D) Dissociation.
10. Which row in the table shows how the metal will react?

	Metal	Reacts with dilute acid	Reacts rapidly with cold water	Reacts with steam
(A)	Copper	✓	✗	✗
(B)	Calcium	✗	✓	✓
(C)	Zinc	✓	✗	✗
(D)	Magnesium	✓	✗	✓

--	--	--	--	--	--	--	--

Student Number

Part B – 25 marks

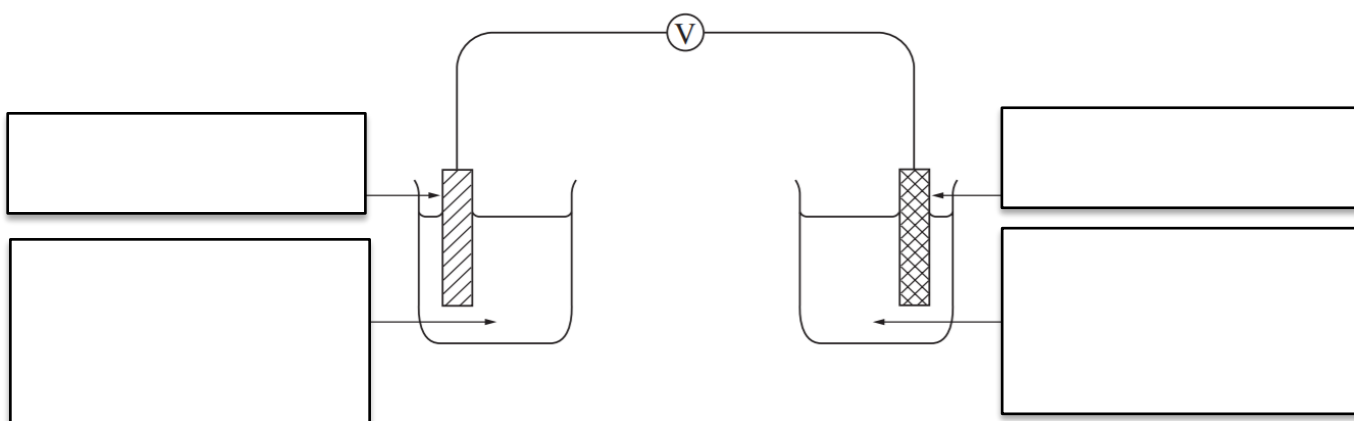
Attempt Questions 11 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (5 marks)

The galvanic cell below is made by connecting two half-cells. One half-cell was made by putting an aluminium electrode in an aluminium nitrate solution. The other half-cell was made by putting a lead electrode in a lead nitrate solution. The electrodes were connected to a voltmeter as shown in the diagram.



Add the following labels to the diagram:

3P

- Which metal is the Anode and cathode
- Direction of electron flow
- Direction of anion and cation flow
- The two nitrate solutions
- A salt-bridge

- a) Using the standard potentials table in the data sheet, calculate the theoretical voltage of this galvanic cell.

2K

.....

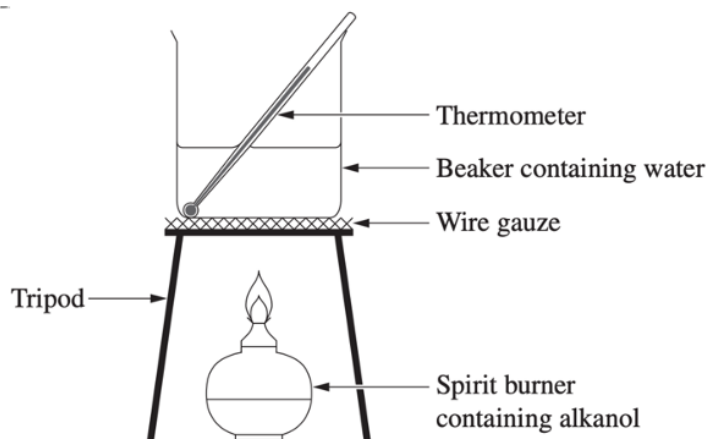
Question 12 (7 marks)

Three groups of students set out to determine the heats of combustion of the following alcohols:

- Methanol, CH_3OH
- Ethanol, $\text{C}_2\text{H}_5\text{OH}$
- Propan-1-ol, $\text{C}_3\text{H}_7\text{OH}$

Each group of students measured out 100mL of water into a container and heated the water by burning a measured mass of each of the alcohols.

Their results are shown in the table below:



Fuel (alcohol)	Amount of water heated (g)	Temperature rise (K)	Mass of alcohol burnt (g)	Heat of combustion ($\text{kJ}\cdot\text{mol}^{-1}$)
Methanol	100	10	0.185	725
Ethanol	100	10	0.142	?
Propan-1-ol	100	10	0.125	2016

- a) From the data in the table, calculate the molar heat of combustion of ethanol. 4K

.....

.....

.....

.....

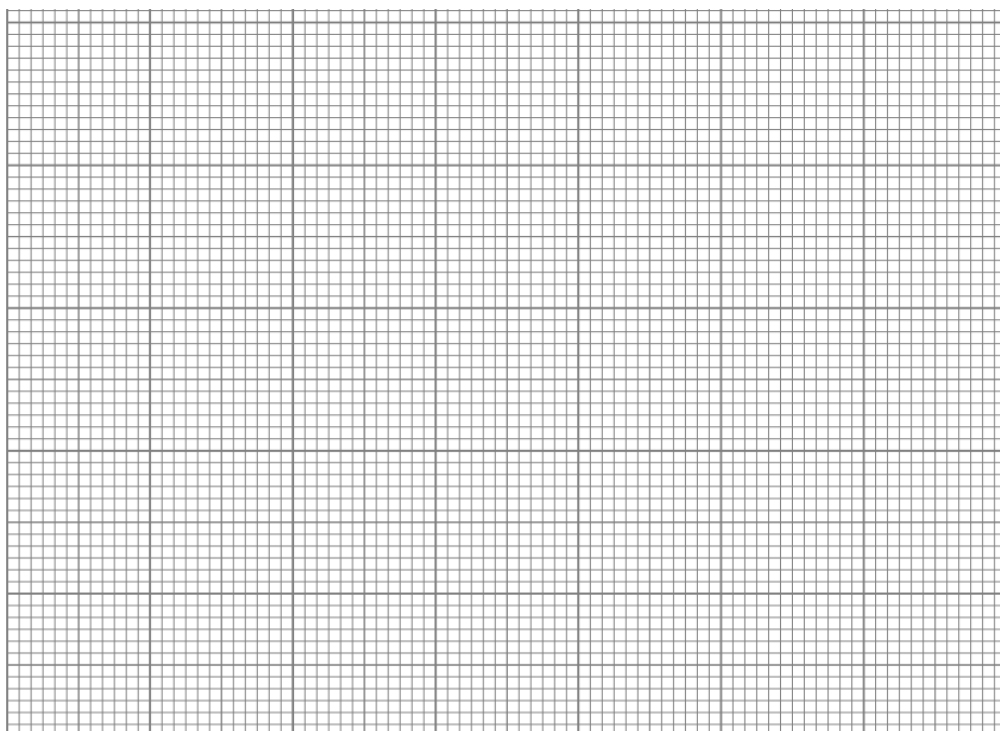
.....

.....

.....

.....

- b) On the grid below, draw a graph representing the molar heat of combustion for the three alcohols against the molar mass of the alcohols. 3A



--	--	--	--	--	--	--	--

Student Number

Part C – 13 marks

Attempt Questions 13 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 12 (5 marks)

- a) Write a method to describe an experiment you performed in a school laboratory to measure the change in reaction rate with concentration. 3P

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- b) Using collision theory, explain why increasing the temperature increases the rate of any chemical reaction. 2K

.....

.....

.....

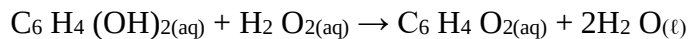
.....

.....

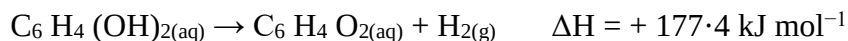
.....

Question 14 (4 marks)

The overall equation representing the formation of quinone, $\text{C}_6\text{H}_4\text{O}_2$ is as follows:



Use the following data to calculate the enthalpy change, in kJ mol^{-1} , for the above reaction. 4A



.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

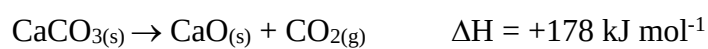
.....

Question 15 (4 marks)

Use the data in the table to answer this question.

	CaCO_{3(s)}	CaO_(s)	CO_{2(g)}
S (J mol⁻¹ K⁻¹)	90	40	214

Calcium carbonate decomposes according to the following equation at 25°C.



Calculate ΔG for this reaction.

4K

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and
at 0°C (273.15 K) 22.71 L
at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

KEY

Atomic Number		Symbol		Name	
79		Au		Gold	
197.0					

Standard Atomic Weight		Name	
197.0		Gold	

1	H	1.008	Hydrogen	3	Li	6.941	Lithium	4	Be	9.012	Beryllium	12	Mg	24.31	Magnesium	20	Ca	40.08	Calcium	38	Sr	87.62	Strontium	56	Ba	137.3	Barium	88	Ra	226	Radium	87	Fr	223	Francium
2	He	4.003	Helium	10	Ne	20.18	Neon	18	Ar	39.95	Argon	36	Kr	83.80	Krypton	54	Xe	131.3	Xenon	86	Rn														
5	B	10.81	Boron	13	Al	26.98	Aluminum	31	Ga	69.72	Gallium	49	In	114.8	Indium	81	Tl	204.4	Thallium	113	Nh														
6	C	12.01	Carbon	14	Si	28.09	Silicon	32	Ge	72.64	Germanium	50	Sn	118.7	Tin	82	Pb	207.2	Lead	114	Fl														
7	N	14.01	Nitrogen	15	P	30.97	Phosphorus	33	As	74.92	Arsenic	51	Sb	121.8	Antimony	83	Bi	209.0	Bismuth	115	Mc														
8	O	16.00	Oxygen	16	S	32.07	Sulfur	34	Se	78.96	Selenium	52	Te	127.6	Tellurium	84	Po		Polonium	116	Lv														
9	F	19.00	Fluorine	17	Cl	35.45	Chlorine	35	Br	79.90	Bromine	53	I	126.9	Iodine	85	At		Astatine	117	Ts														
10	Ne	20.18	Neon	18	Ar	39.95	Argon	36	Kr	83.80	Krypton	54	Xe	131.3	Xenon	86	Rn		Radon	118	Og														

Lanthanoids																Actinoids																																																																																															
57	La	138.9	Lanthanum	59	Pr	140.9	Praseodymium	61	Pm		Promethium	63	Eu	152.0	Europium	65	Gd	157.3	Gadolinium	67	Ho	164.9	Holmium	69	Tm	168.9	Thulium	71	Lu	175.0	Lutetium	89	Ac	227	Actinium	91	Pa	231.0	Protactinium	93	Np	237.0	Neptunium	95	Am	243	Americium	97	Bk	247	Berkelium	99	Es	252	Einsteinium	101	Md	288	Mendelevium	103	Lr	262	Lawrencium																																																
58	Ce	140.1	Cerium	60	Nd	144.2	Niodymium	62	Sm	150.4	Samarium	64	Gd	157.3	Gadolinium	66	Dy	162.5	Dysprosium	68	Er	167.3	Erbium	70	Yb	173.1	Ytterbium	72	Hf	178.5	Hafnium	80	Zr	91.22	Zirconium	82	Hf	178.5	Hafnium	84	Rf	104	Rutherfordium	86	Db	105	Dubnium	88	Ra	226	Radium	90	Th	232.0	Thorium	92	U	238.0	Uranium	94	Pu	244	Plutonium	96	Cm	247	Curium	98	Cf	251	Californium	100	Fm	257	Fermium	102	No	289	Nobelium	104	Rf	261	Rutherfordium	106	Sg	266	Seaborgium	108	Hs	277	Hassium	110	Ds	271	Darmstadtium	112	Cn	285	Copernicium	114	Fl	289	Flerovium	116	Lv	293	Livermorium	118	Og	294	Oganesson

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.



Sydney Girls High School

--	--	--	--	--	--	--	--

Student Number

2023

YEAR 11

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided on pages 13 -15
- Write your student number on the multiple choice answer sheet and the cover page

Total marks – 35

Part A – Pages 2-5

10 marks

- Attempt Questions 1 – 10

Part B – Pages 6-8

13 marks

- Attempt Questions 11 – 12

Part B – Pages 9-11

12 marks

- Attempt Questions 13 – 15

Part A – 10 marks

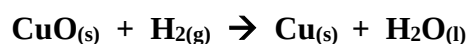
Attempt Questions 1 – 10

Use the multiple-choice answer sheet for Questions 1–10

- 1 Which observation will be made when a clean strip of magnesium is dipped into aqueous copper sulfate:

- (A) Hydrogen gas is released.
- (B) Sulfur dioxide gas is released.
- (C) The blue colour of the copper nitrate becomes more intense.
- (D) The magnesium strip becomes coated with a reddish deposit of copper.

- 2 Hydrogen and copper (II) oxide react to produce metal in water.



Which row correctly describes the oxidising and reducing agent?

	Oxidising agent	Reducing agent
(A)	H ₂ O	Cu
(B)	H ₂	CuO
(C)	Cu	H ₂ O
(D)	CuO	H ₂

- 3 Which combination of the following periodic table trends would result in an increase in a metal's reactivity?

- (A) Decrease in atomic radius, decrease in electronegativity.
- (B) Increase in ionisation energy, increase in atomic radius.
- (C) Increase in ionisation energy, decrease in electronegativity.
- (D) Decrease in ionisation energy, decrease in electronegativity.

4 Which equation represents a neutralisation reaction?

- (A) $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(l)}$
- (B) $\text{C}_6\text{H}_{12}\text{O}_{6(aq)} + 6\text{O}_{2(g)} \rightarrow 6\text{CO}_{2(g)} + 6\text{H}_2\text{O}_{(g)}$
- (C) $\text{CuCl}_{2(aq)} + 2\text{NaOH}_{(aq)} \rightarrow \text{Cu}(\text{OH})_{2(s)} + 2\text{NaCl}_{(aq)}$
- (D) $2\text{HNO}_{3(l)} + \text{Ba}(\text{OH})_{2(aq)} \rightarrow \text{Ba}(\text{NO}_3)_{2(aq)} + 2\text{H}_2\text{O}_{(l)}$

5 Which of the following is a true statement that can be used to predict the entropy change in a system?

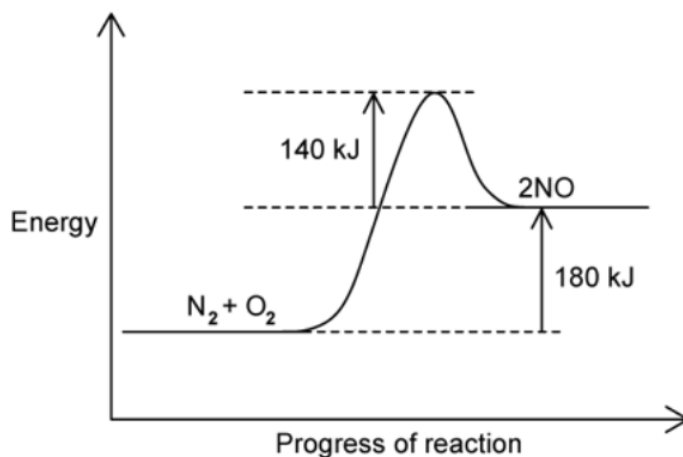
- (A) As the temperature of a substance increase, the entropy increases.
- (B) A substance in its gaseous state has less entropy, than the same amount of the substance in its liquid state.
- (C) A substance in its liquid state has less entropy than the same amount of substance in its solid state.
- (D) When a solute dissolves in a solvent, the solute particles in solution can be arranged in fewer ways than when they were solid, so the amount of entropy usually decreases

6 Catalysts such as iron and nickel are commonly used in industrial reactions.

Which statement about a catalyst is NOT correct?

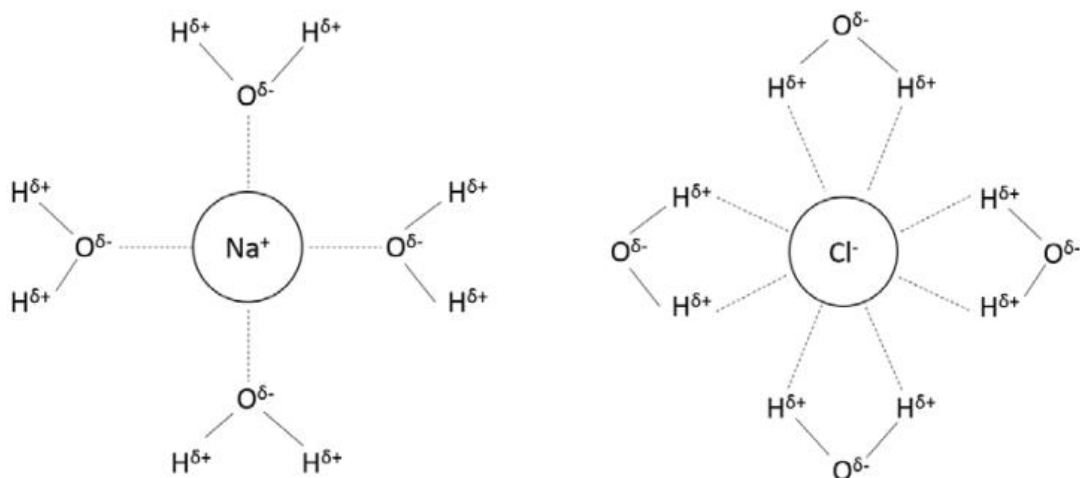
- (A) They create a pathway which has a lower activation energy.
- (B) Their concentration does not change throughout the reaction.
- (C) They increase the amount of energy required for a reaction to occur.
- (D) They can allow reaction to occur at much lower temperatures.

- 7 What is the activation energy for this reaction?



- (A) +140 kJ
- (B) +180 kJ
- (C) +320 kJ
- (D) -320 kJ
- 8 Which of the following is a statement describing Hess' Law?
- (A) Energy cannot be created nor destroyed, but it can be changed from one form into another.
- (B) In a chemical reaction, the total mass of the reactants is equal to the total mass of the products.
- (C) The enthalpy change for a chemical reaction is the same regardless of the number of steps involved in the process.
- (D) Enthalpy change in a chemical reaction equals the sum of the enthalpy of all products minus the sum of enthalpies of all reactants.

9 Diagram and models are often used to demonstrate complex chemical ideas.



Which chemical process is demonstrated by these diagrams?

- (A) Ionisation.
- (B) Carbonation.
- (C) Precipitation.
- (D) Dissociation.

10. Which row in the table shows how the metal will react?

	Metal	Reacts with dilute acid	Reacts rapidly with cold water	Reacts with steam
(A)	Copper	✓	✗	✗
(B)	Calcium	✗	✓	✓
(C)	Zinc	✓	✗	✗
(D)	Magnesium	✓	✗	✓

--	--	--	--	--	--	--	--

Student Number

Part B – 25 marks

Attempt Questions 11 - 15

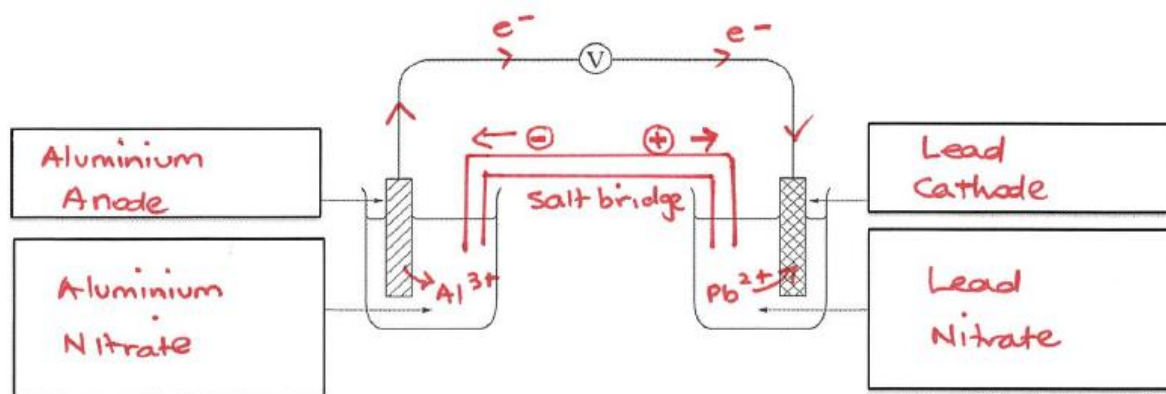
Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (5 marks)

The galvanic cell below is made by connecting two half-cells. One half-cell was made by putting an aluminium electrode in an aluminium nitrate solution. The other half-cell was made by putting a lead electrode in a lead nitrate solution. The electrodes were connected to a voltmeter as shown in the diagram.

Correct labels	3
Substantially correct labels	2
Some relevant labels	1



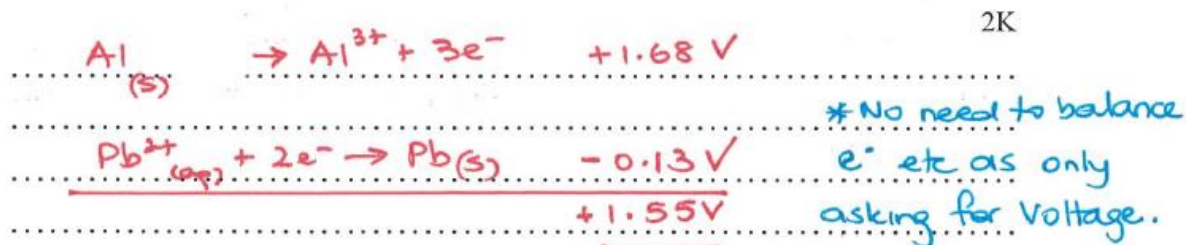
Add the following labels to the diagram:

3P

- Which metal is the Anode and Cathode
- Direction of electron flow
- Direction of anion and cation flow
- The two nitrate solutions
- A salt-bridge

- a) Using the standard potentials table in the data sheet, calculate the theoretical voltage of this galvanic cell. 2K

Correct Calculation	2
One relevant step in calculation	1



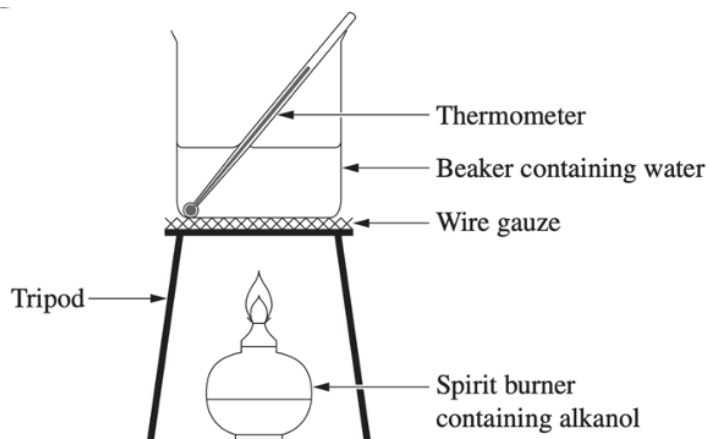
Question 12 (7 marks)

Three groups of students set out to determine the heats of combustion of the following alcohols:

- Methanol, CH₃OH
- Ethanol, C₂H₅OH
- Propan-1-ol, C₃H₇OH

Each group of students measured out 100mL of water into a container and heated the water by burning a measured mass of each of the alcohols.

Their results are shown in the table below:



Fuel (alcohol)	Amount of water heated (g)	Temperature rise (K)	Mass of alcohol burnt (g)	Heat of combustion (kJ.mol ⁻¹)
Methanol	100	10	0.185	725
Ethanol	100	10	0.142	?
Propan-1-ol	100	10	0.125	2016

a) From the data in the table, calculate the molar heat of combustion of ethanol. 4K

Any two parts of calculation performed eg • Calculation of heat supplied (eg q) • Correct calculation molar mass • Calculation of moles of ethanol • Calculation of ΔH from q (including – ve sign for exo rxn)	4
Most steps correct	3
Some relevant calculations	2
One relevant calculation	1

$$\begin{aligned}
 q &= mc\Delta T \\
 &= (100)(4.18)(10) \\
 \therefore q &= 4180 \text{ J or } 4.180 \text{ kJ} \\
 n &= m/M \\
 &= 0.142/46.068 \\
 \therefore n &= 0.0031 \text{ mol} \\
 \Delta H &= -q/n \\
 &= -4.180/0.0031 \\
 \therefore \Delta H &= -1348 \text{ kJ/mol}
 \end{aligned}$$

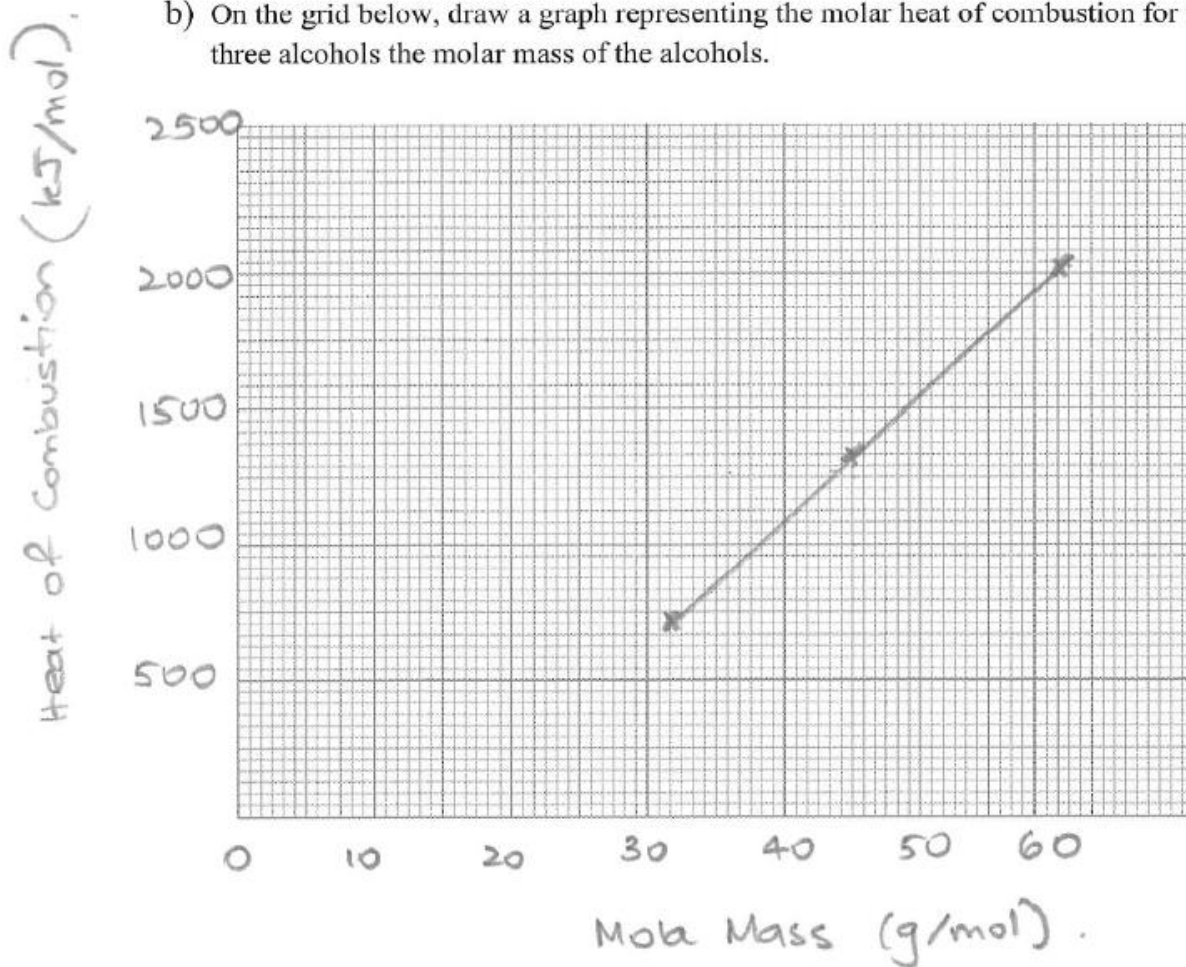
Graphed correctly including:

3

• Labels of both axes with correct units • Using > 50% of grid (appropriate linear scale used) • Correct plotting of points • Curve straight line	
As above but with 1 omission/error	2
As above but with 2 omissions/errors	1

b) On the grid below, draw a graph representing the molar heat of combustion for the three alcohols the molar mass of the alcohols.

3A



--	--	--	--	--	--	--	--

Student Number

Part C – 13 marks

Attempt Questions 13 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 12 (5 marks)

- a) Write a method to describe an experiment you performed in a school laboratory to measure the change in reaction rate with concentration.

3P

Correct description of the method	3
Substantially correct description of the method	2
Some relevant steps	1

1. Measure + Place 50 mL of $\text{Na}_2\text{S}_2\text{O}_3$ into a 250 mL conical flask
2. Place the flask over a cross marked on white paper
3. Measure 5 mL of 1M HCl, add to the $\text{Na}_2\text{S}_2\text{O}_3$ in the conical flask; start the stopwatch
4. Stop the watch and record the time taken for the cross to become invisible when viewed from above
5. Record your results in a data table.
6. Repeat procedure using 40 mL, 30 mL, 20 mL, 10 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution made up to a total volume of 50 mL with distilled water.

- b) Using collision theory, explain why increasing the temperature increases the rate of any chemical reaction.

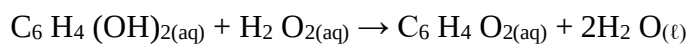
2K

Correct explanation	2
Partially correct explanation	1

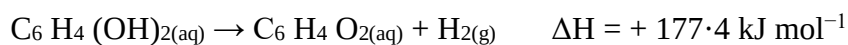
In collision theory, temp affects the rate of collision as well as the energy of the molecules. Higher temps $\uparrow$ the KE of the molecules so more molecules have the minimum KE (or $\geq$ than E_a). This increases the rate of the reaction.

Question 14 (4 marks)

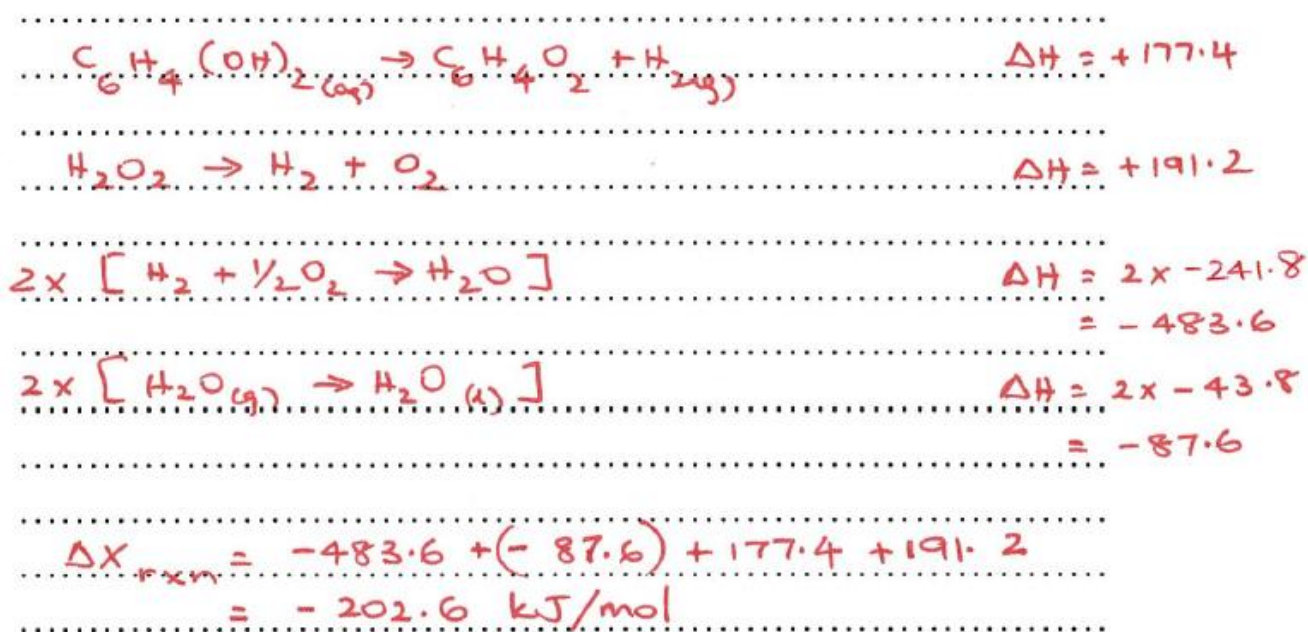
The overall equation representing the formation of quinone, $C_6H_4O_2$ is as follows:



Use the following data to calculate the enthalpy change, in kJ mol^{-1} , for the above reaction. 4A



All correct calculations, including significant figures	4
All steps correct	3
Several relevant calculations	2
Some relevant information	1



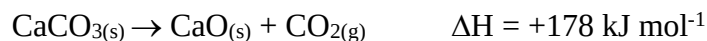
Question 15 (4 marks)

Use the data in the table to answer this question.

	$\text{CaCO}_{3(\text{s})}$	$\text{CaO}_{(\text{s})}$	$\text{CO}_{2(\text{g})}$
--	-----------------------------	---------------------------	---------------------------

S (J mol ⁻¹ K ⁻¹)	90	40	214
--	----	----	-----

Calcium carbonate decomposes according to the following equation at 25°C.



Calculate ΔG for this reaction.

4K

All correct calculations	4
Most steps correct	3
Some relevant calculations	2
One relevant calculation	1

$$\begin{aligned} \Delta S &= \text{products} - \text{reactants} \\ &= (40 + 214) - 90 \\ \therefore \Delta S &= +164 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= +0.164 \text{ kJ mol}^{-1} \text{ K}^{-1} \\ \Delta H &= +178 \text{ kJ mol}^{-1} \\ T &= 298.15 \text{ K} \\ \Delta G &= +178 - (298.15 \times 0.164) \\ &= +129.103 \text{ kJ} \\ \therefore \Delta G &= +129 \text{ kJ} \quad (2 \text{ sig figs}). \end{aligned}$$

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and
at 0°C (273.15 K) 22.71 L
at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1	2	KEY																3	4																																																																																																																																																																																																																																																										
H 1.008 Hydrogen	He 4.003 Helium																			Li 6.941 Lithium	Be 9.012 Beryllium																			B 10.81 Boron	C 12.01 Carbon	N 14.01 Nitrogen	O 16.00 Oxygen	F 19.00 Fluorine	Ne 20.18 Neon																																																																																																																																																																																																																																
11	Na 22.99 Sodium	12	Mg 24.31 Magnesium																	13	Al 26.98 Aluminium	14	Si 28.09 Silicon	15	P 30.97 Phosphorus	16	S 32.07 Sulfur	17	Cl 35.45 Chlorine	18	Ar 39.95 Argon																	19	K 39.10 Potassium	20	Ca 40.08 Calcium	21	Sc 44.96 Scandium	22	Ti 47.87 Titanium	23	V 50.94 Vanadium	24	Cr 52.00 Chromium	25	Mn 54.94 Manganese	26	Fe 55.85 Iron	27	Co 58.93 Cobalt	28	Ni 58.69 Nickel	29	Cu 63.55 Copper	30	Zn 65.38 Zinc	31	Ga 69.72 Gallium	32	Ge 72.64 Germanium	33	As 74.92 Arsenic	34	Se 78.96 Selenium	35	Br 79.90 Bromine	36	Kr 83.80 Krypton																	37	Rb 85.47 Rubidium	38	Sr 87.61 Strontium	39	Y 88.91 Yttrium	40	Zr 91.22 Zirconium	41	Nb 92.91 Niobium	42	Mo 95.96 Molybdenum	43	Tc 99.96 Technetium	44	Ru 101.1 Ruthenium	45	Rh 102.9 Rhodium	46	Pd 106.4 Palladium	47	Ag 107.9 Silver	48	Cd 112.4 Cadmium	49	In 114.8 Indium	50	Sn 118.7 Tin	51	Sb 121.8 Antimony	52	Te 127.6 Tellurium	53	I 126.9 Iodine	54	Xe 131.3 Xenon																	55	Cs 132.9 Cesium	56	Ba 137.3 Barium	57-71	Lanthanoids																72	Hf 178.5 Hafnium	73	Ta 180.9 Tantalum	74	W 183.9 Tungsten	75	Re 186.2 Rhenium	76	Os 190.2 Osmium	77	Ir 192.2 Iridium	78	Pt 195.1 Platinum	79	Au 197.0 Gold	80	Hg 200.6 Mercury	81	Tl 204.4 Thallium	82	Pb 207.2 Lead	83	Bi 209.0 Bismuth	84	Po 209.0 Polonium	85	At 209.0 Astatine	86	Rn 209.0 Radon																	87	Fr 223.0 Francium	88	Ra 226.0 Radium	89-103	Actinoids																104	Rf 261.1 Rutherfordium	105	Db 262.1 Dubnium	106	Sg 266.1 Seaborgium	107	Bh 264.1 Berkelium	108	Hs 277.1 Hassium	109	Mt 268.1 Meitnerium	110	Ds 271.1 Darmstadtium	111	Rg 272.1 Roentgenium	112	Cn 285.1 Copernicium	113	Nh 284.1 Nihonium	114	Fl 289.1 Flerovium	115	Mc 288.1 Moscovium	116	Lv 293.1 Livermorium	117	Ts 294.1 Tennessine	118	Og 294.1 Oganesson

Lanthanoids

57	La	Lanthanum	58	Ce	Cerium	59	Pr	Praseodymium	60	Nd	Neodymium	61	Pm	Promethium	62	Sm	Samarium	63	Eu	Europium	64	Gd	Gadolinium	65	Tb	Terbium	66	Dy	Dysprosium	67	Ho	Holmium	68	Er	Erbium	69	Tm	Thulium	70	Yb	Ytterbium	71	Lu	Lutetium
----	----	-----------	----	----	--------	----	----	--------------	----	----	-----------	----	----	------------	----	----	----------	----	----	----------	----	----	------------	----	----	---------	----	----	------------	----	----	---------	----	----	--------	----	----	---------	----	----	-----------	----	----	----------

Actinoids

89	Ac	Actinium	90	Th	Thorium	91	Pa	Protactinium	92	U	Uranium	93	Np	Neptunium	94	Pu	Plutonium	95	Am	Americium	96	Cm	Curium	97	Bk	Berkelium	98	Cf	Californium	99	Es	Einsteinium	100	Fm	Fermium	101	Md	Mendelevium	102	No	Nobelium	103	Lr	Lawrencium
----	----	----------	----	----	---------	----	----	--------------	----	---	---------	----	----	-----------	----	----	-----------	----	----	-----------	----	----	--------	----	----	-----------	----	----	-------------	----	----	-------------	-----	----	---------	-----	----	-------------	-----	----	----------	-----	----	------------

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

